



Configuring match and merge

Matching policy



Entity Type

	 Basic	 ID-Matched	 Fuzzy-matched
Publishers	One or none	Multiple	Multiple
Match and Merge	Not supported Dups detected at creation time	Yes , using exact match on a shared ID Dups detected at creation time	Yes , using match rules Dups detected at creation time
Survivorship	Not supported	Consolidation is supported Value override is supported	Consolidation is supported Value override is supported
ID Management	Golden ID=Source ID	All source systems use a shared ID Golden ID=Source ID	Source systems have different IDs The Golden ID is generated
Recommended for	Centralized pattern Single source system Reference Data	Consolidation pattern, for entities with a shared ID.	Consolidation pattern, for entities with no shared ID.
Example	Product Family, Cost Centers	Products (with a shared SKU)	Customers, Contacts, Products (with no shared SKU)

Matching process

- **Detects duplicates** to consolidate them into a **golden record**
- **ID Matching**: Records are **exact** matched on the **primary key attribute**
- **Fuzzy Matching**: Records are matched by detecting similar values in the enriched and validated records.
- **Basic entities do not support matching and consolidation**

Matching during authoring

ID and **Basic** entities **can** have match rules

To detect potential **duplicates** during **authoring**

Can be used within workflows and steppers

You can configure what happens when duplicates are detected during authoring

Overview

1. Matching detects **duplicate pairs** (similar pairs of records):
 - Multiple **match rules** in a matcher allow you to define **several conditions** for considering two records a match
 - Each condition has a different **match score**
 - When two records match, they receive a **match score equal** to the **highest score of all the rules** that matched them (highest confidence).
2. **Match groups** are created from the duplicate pairs:
 - Uses matching transitivity - if A matches B and B matches C, then A, B and C are in the same matching group
 - Each match group has a **confidence score**

Match rule expression

A **match rule** is a SemQL expression evaluating two records, record1 and record2:

For example:

- Simplest possible exact match rules:

```
Record1.Email = Record2.Email
```

```
Record1.CustomerName = Record2.CustomerName
```

- Fuzzy match:

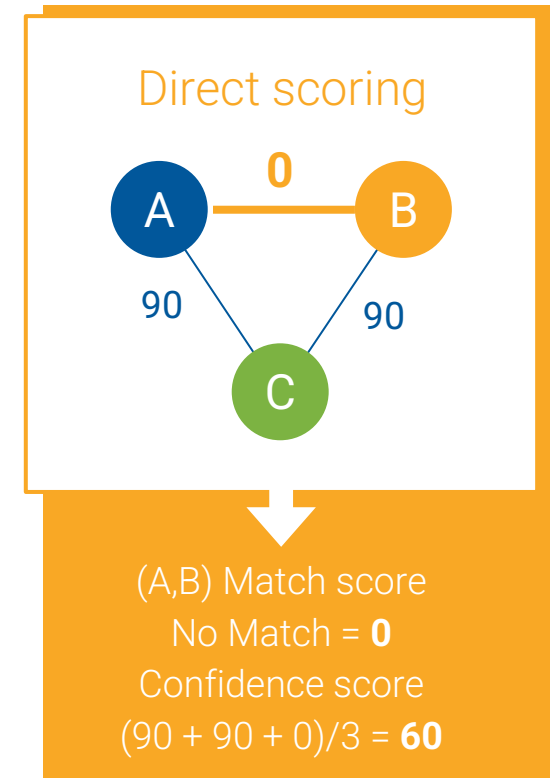
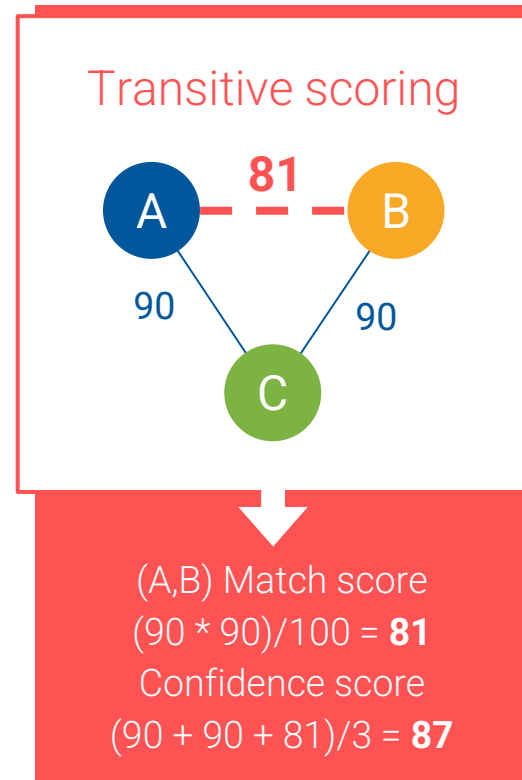
```
SEM_EDIT_DISTANCE_SIMILARITY( Record1.CustomerName,  
Record2.CustomerName ) > 85
```

Match rule examples

	Score	Expression
Exact match on name and address	100	<pre>Record1.NormalizedCompanyName = Record2.NormalizedCompanyName AND Record1.InputAddress.AddressLine = Record2.InputAddress.AddressLine AND Record1.InputAddress.City = Record2.InputAddress.City AND (Record1.InputAddress.State = Record2.InputAddress.State OR Record1.InputAddress.State IS NULL OR Record2.InputAddress.State IS NULL) AND Record1.InputAddress.ZipCode = Record2.InputAddress.ZipCode AND Record1.InputAddress.Country = Record2.InputAddress.Country</pre>
Phonetic match on name	88	<pre>Record1.PhoneticCompanyName = Record2.PhoneticCompanyName AND substr(coalesce(Record1.GeocodedAddress.PostalCode, Record1.InputAddress.ZipCode), 1, 5) = substr(coalesce(Record2.GeocodedAddress.PostalCode, Record2.InputAddress.ZipCode), 1, 5)</pre>
Fuzzy match on name	76	<pre>SEM_EDIT_DISTANCE(Record1.NormalizedCompanyName, Record2.NormalizedCompanyName) <= 2 AND LENGTH(Record1.NormalizedCompanyName) >= 10</pre>

Transitive scoring

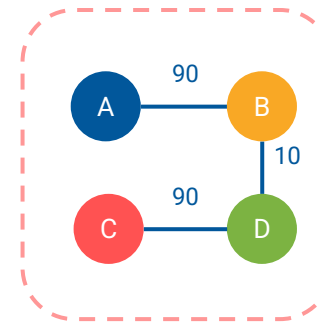
- Default method to score pair of records
- More intuitive way to score:
 - Propagate similarity as probabilities
 - Uses indirect paths
 - Match scores combined as match probabilities
 - Override direct scores if higher
- Transitive Confidence Score
 - Average of Transitive Match Score
- Deactivated for groups of 300+ records



Iterative clustering

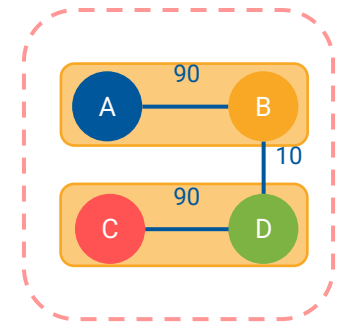
- Optional iterative clustering
- More intuitive way to create golden records
- One iteration per match score
- Create golden step by step instead of once based on the global confidence score

Single pass clustering



Merge Threshold: 80
Confidence score
 $(90 + 90 + 10)/3 \approx 63$
No golden created

Iterative clustering



Golden created for A and B
Golden created for C and D
A/B is suggested with C/D



Please refer to the matching best practices guide for advanced concepts and tips



Configuring match and merge

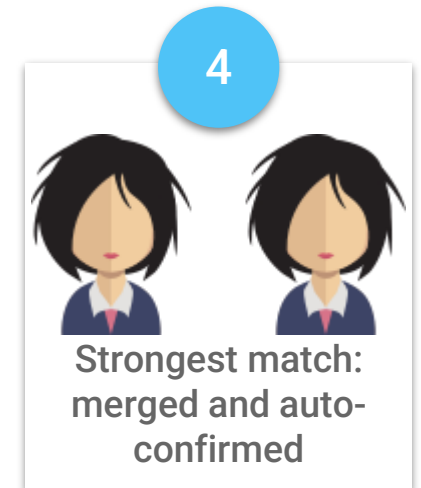
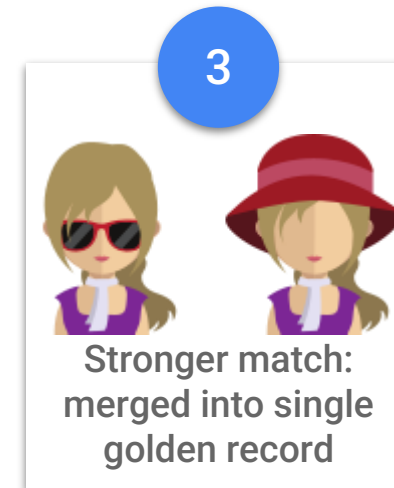
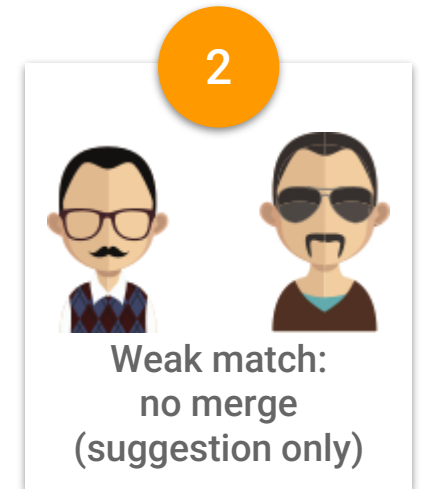
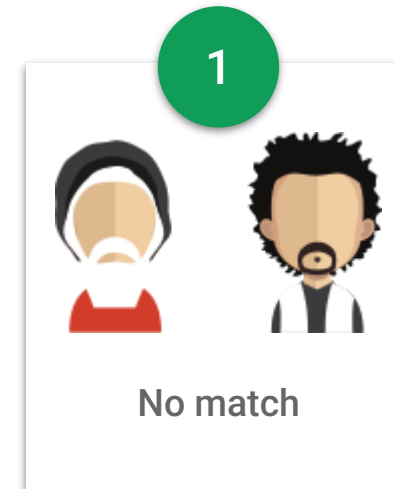
Merge policy



Merge policy

Match rules define how incoming source records will be matched together.

The **merge policy** defines the **thresholds** for whether matches should be **merged** and for whether merges should be **confirmed**.



Merge policy

Defines the **confidence scores** required to **automatically merge groups**

▼ Merge Policy	
Create a golden record from new master records:	70
Merge unconfirmed golden records:	70
Merge confirmed golden records:	90
Merge unconfirmed with confirmed golden records:	90
Add new master records to an unconfirmed golden record:	70
Add new master records to a confirmed golden record:	80
Merge confirmed golden records previously split by the user:	90

▼ Auto-Confirm Policy	
Auto-confirm golden records:	75
Auto-confirm singletons:	☐



More information can be found in the Data Hub Design Guide and the matching best practice guide.

<https://www.semarchy.com/doc/semarchy-xdm/xdm/5.3/Design/matching/matching.html>

Match on Child Entities

- **Matching on child** records is available for **fuzzy** matched entities **only**
 - If you have a **customer** entity with a **child address entity**, you can use its **street** attribute to **match customers** in the match rule
 - E.g.: `Record1.Street = Record2.Street`

Match On:	Addresses
Match On Child Records:	☒
Child Records:	Addresses

- If any pair of child records match according to the rule, then the two parent records are matched